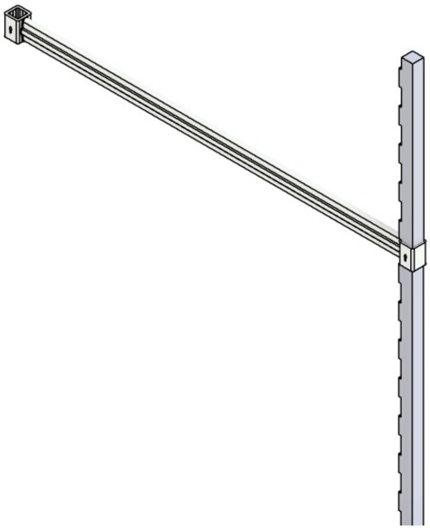
Study Report



Analyzed File	5th test v2	Version	Autodesk Fusion (2604.1.48)
Creation Date	2025-10-24, 20:58:43	Author	User
Summary	5th and final shit		

Report Properties

Report Properties	
Title	Studies
Author	User

Simulation Model 1

Study 1 - Static Stress

Study Properties

Study Properties	
Study Type	Static Stress
Last Modification Date	2025-10-24, 20:49:54

Study Setup

Settings	
General	
Contact Tolerance	0.10 mm
Remove Rigid Body Modes	No
Mesh	
Average Element Size (% of model size) - Solids	10
Scale Mesh Size Per Part	No
Average Element Size (absolute value)	-
Element Order	Parabolic
Create Curved Mesh Elements	Yes
Max. Turn Angle on Curves (Deg.)	60
Max. Adjacent Mesh Size Ratio	1.5
Max. Aspect Ratio	10

Minimum Element Size (% of average size)	20
Adaptive Mesh Refinement	
Number of Refinement Steps	0
Results Convergence Tolerance (%)	20
Portion of Elements to Refine (%)	10
Results for Baseline Accuracy	von Mises Stress

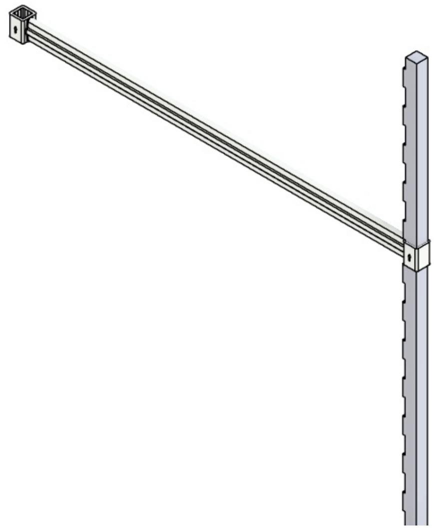
Materials	
Plastic	
Components	
Component	Strength
Plug in play Final v1:1/Final Top lock v1:1	Yield Strength
Properties	
Density	1.290E-06 kg / mm^3
Young's Modulus	709.00 MPa
Poisson's Ratio	0.40
Yield Strength	30.00 MPa
Ultimate Tensile Strength	40.00 MPa
Thermal Conductivity	2.500E-04 W / (mm C)
Thermal Expansion Coefficient	4.190E-05 / C
Specific Heat	1750.00 J / (kg C)
Steel	
Components	
Component	Strength
Small Leg - Plug final v1:1/Smalle-Leg-Finalr v1 (1):1	Yield Strength
Smalle-Leg-Finalr v1:1	Yield Strength
Properties	
Density	7.850E-06 kg / mm^3
Young's Modulus	210000.00 MPa
Poisson's Ratio	0.30
Yield Strength	207.00 MPa
Ultimate Tensile Strength	345.00 MPa
Thermal Conductivity	0.056 W / (mm C)
Thermal Expansion Coefficient	1.200E-05 / C
Specific Heat	480.00 J / (kg C)

Contacts				
Contact Type	Contact Set	Penetration Type	Bodies	Entities
Bonded				
	Bonded1	Symmetric	[Body1 Final Top lock v1:1||Body1 Smalle-Leg-Finalr v1 (1):1]	[Face 39||Face 5]
	[M] Bonded2	Symmetric	[Body1 Smalle-Leg-Finalr v1 (1):1||Body1 Smalle-Leg-Finalr v1:1]	[Face 1||Face 328]

Mesh	
Type	Solids
Nodes	286434
Elements	180236

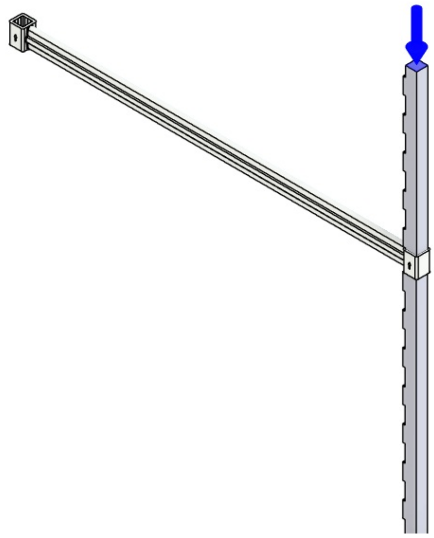
Load Case1

Constraints
Fixed1



Type	Fixed
Ux	Fixed
Uy	Fixed
Uz	Fixed

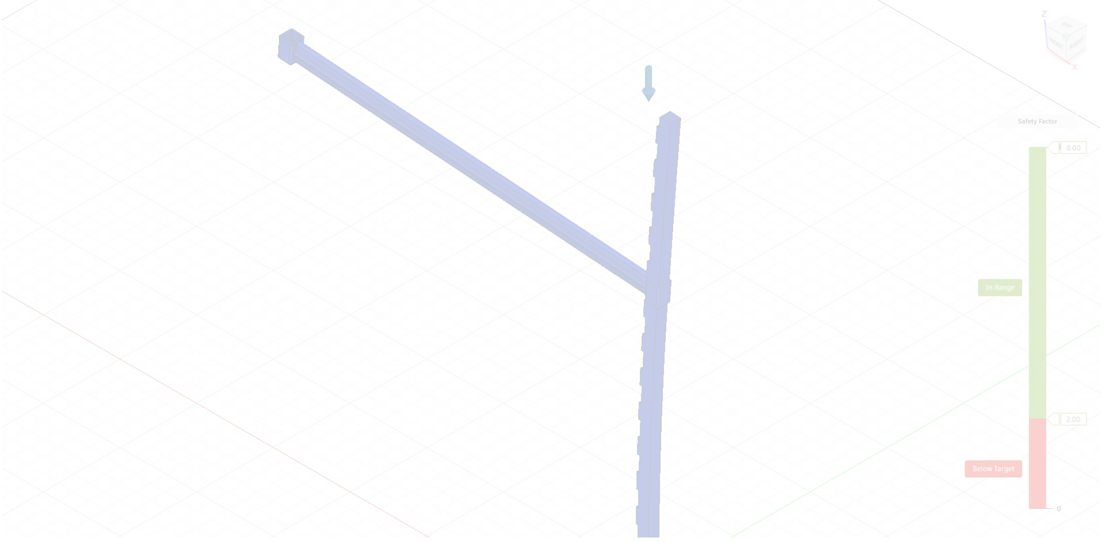
Loads
Force1



Type	Force
Magnitude	4000.00 N
X Value	0.00 N
Y Value	0.00 N
Z Value	-4000.00 N
Force Per Entity	No

Load Case1 - Guided Results

Safety Factor



With the analysis criteria in this study setup, the design is expected to bend permanently or break.

Next Steps

Below Safety Factor Target

Validate your study setup
- Check the loads are not set too high
- Check the mesh refinement and consider increasing it in areas of high stress
- Confirm your Safety Factor requirement
- If the Displacement result shows high displacement, consider running a Nonlinear study to gain more insight

Check other results
- Examine the Below target Result visualization to see where the weakest areas are
- Check the stress result to see if, and where, the material might yield
- Check the displacement result to see how far from its original alignment the model has moved
- Look at other results, such as reaction forces to check your study set up, or strain to check your material performance

Strengthen weak areas
- Add more material to weak areas
- Select a material with a higher yield strength
- Stiffen the design
- Modify the shape with a Generative Design study

Generate alternative designs
- Add more material to weak areas
- Select a material with a higher yield strength
- Stiffen the design
- Modify the shape with a Generative Design study

Above Safety Factor Limit

Validate your study setup
- Check the loads are not set too high
- Check the mesh refinement and consider increasing it in areas of high stress
- Confirm your Safety Factor requirement
- If the Displacement result shows high displacement, consider running a Nonlinear study to gain more insight

Check other results
- Examine the Below target Result visualization to see where the weakest areas are
- Check the stress result to see if, and where, the material might yield
- Check the displacement result to see how far from its original alignment the model has moved
- Look at other results, such as reaction forces to check your study set up, or strain to check your material performance

Lightweight the design
- Set a Safety Factor upper limit to find where material can be removed
- Remove material from areas that exceed the Safety Factor upper limit

Generate alternative designs
- In the Generative Design workspace, create design modifications
- Remove unnecessary material using a Shape Optimization study

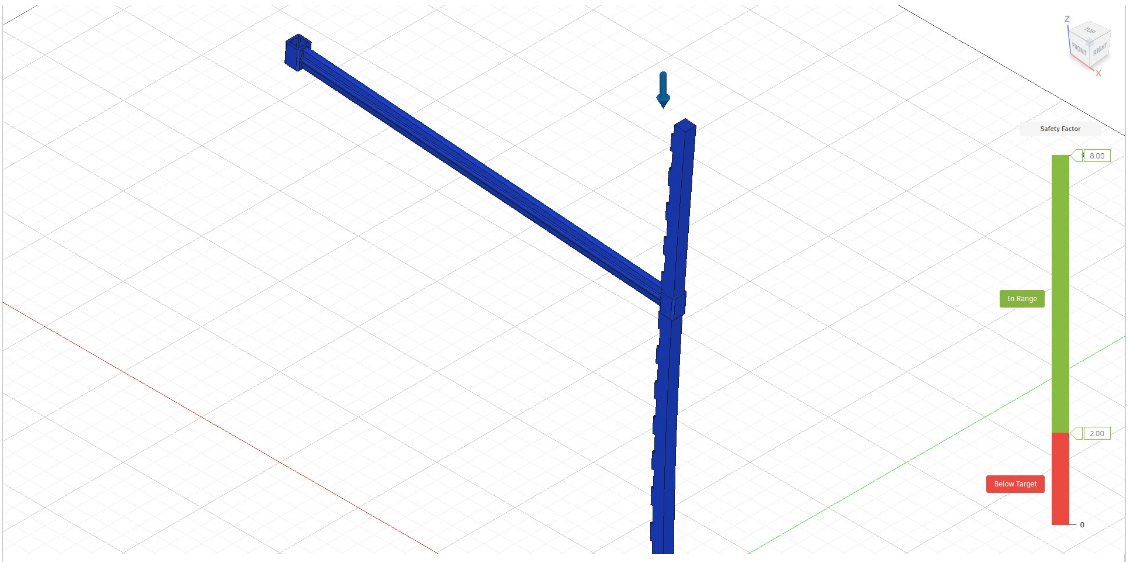
Load Case1 - Results

Result Summary			
Result type	Sub-result	Minimum	Maximum
Safety Factor			
	Safety Factor	15.754	1.078E+11
Stress			

	Von Mises	2.783E-10 MPa	13.14 MPa
	1st Principal	−4.162 MPa	3.275 MPa
	3rd Principal	−16.386 MPa	0.414 MPa
	Normal XX	−9.997 MPa	0.977 MPa
	Normal YY	−4.555 MPa	1.141 MPa
	Normal ZZ	−14.264 MPa	3.193 MPa
	Shear XY	−1.719 MPa	1.685 MPa
	Shear YZ	−2.845 MPa	3.046 MPa
	Shear XZ	−6.804 MPa	6.652 MPa
Displacement			
	Total	0 mm	0.26 mm
	X	−6.334E-05 mm	0.205 mm
	Y	−0.006 mm	0.005 mm
	Z	−0.027 mm	0.243 mm
Reaction Force			
	Total	0 N	1,178.957 N
	X	−160.547 N	129.712 N
	Y	−170.64 N	165.646 N
	Z	0 N	1,178.804 N
Strain			
	Equivalent	7.889E-13	1.491E-04
	1st Principal	−2.711E-12	1.212E-04
	3rd Principal	−1.408E-04	3.49E-12
	Normal XX	−3.327E-05	1.369E-05
	Normal YY	−1.413E-05	1.552E-05
	Normal ZZ	−5.661E-05	2.432E-05
	Shear XY	−4.084E-05	4.916E-05
	Shear YZ	−6.169E-05	8.02E-05
	Shear XZ	−1.207E-04	1.103E-04
Contact Pressure			
	Total	0 MPa	11.385 MPa
	X	−1.394 MPa	3.306 MPa
	Y	−1.205 MPa	3.037 MPa
	Z	−10.463 MPa	5.136 MPa
Contact Force			
	Total	0 N	1,014.776 N
	X	−143.161 N	186.264 N
	Y	−96.731 N	73.76 N
	Z	−798.664 N	1,004.517 N

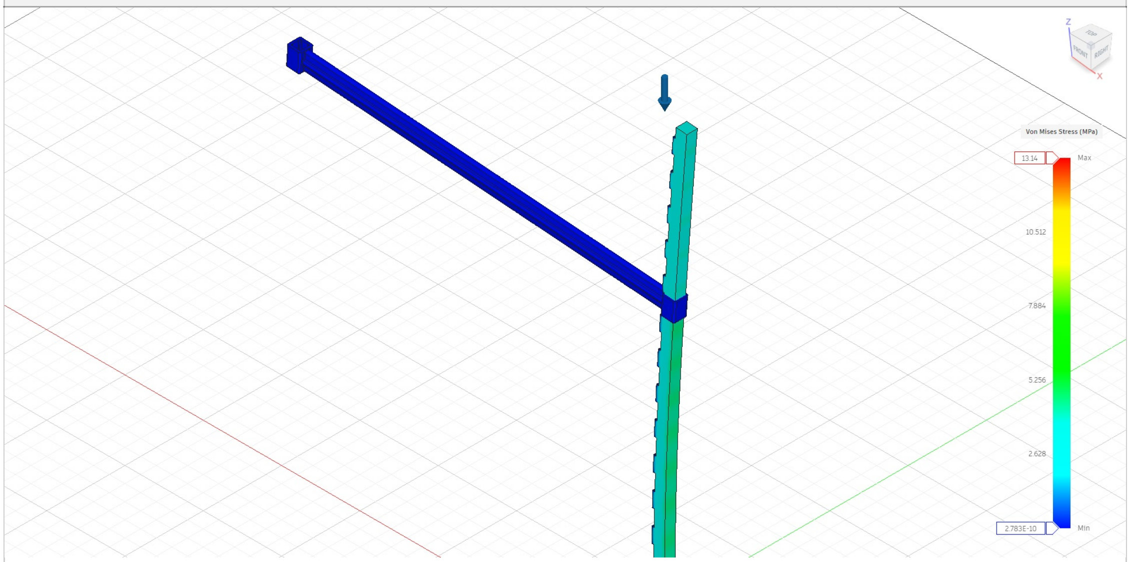
Safety Factor

Safety Factor

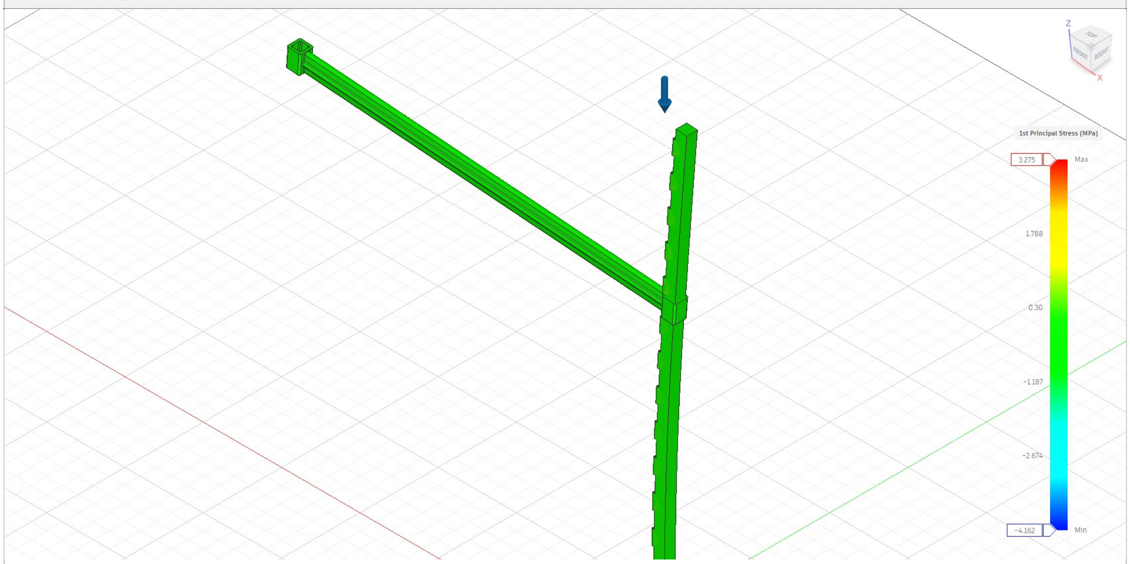


Stress

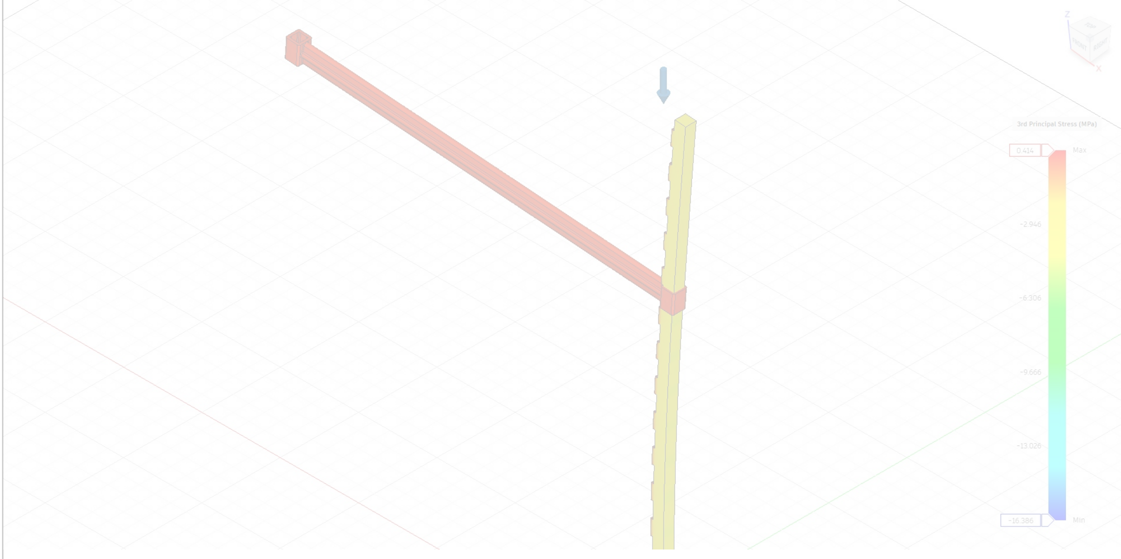
Von Mises



1st Principal



3rd Principal



Displacement

Total